

DEEPLY FELT AFFECT

THE EMERGENCE OF VALENCE IN DEEP ACTIVE INFERENCE



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Partially Observable Markov Decision Processes (POMDPs)



Generative Model for Bayesian Inference

A common framework for inference over discretized states, enabling agents to model complex environments.



Belief Updating & Hidden States

POMDPs update beliefs about hidden states of the world “out there” (s), based on incoming data.

(s) Hidden States: The underlying reality of the world.

(o) Observations: Sensory inputs or outcomes used for updating.

The focus of this framework is the interaction between hidden states and sensory outcomes within discretized state spaces.

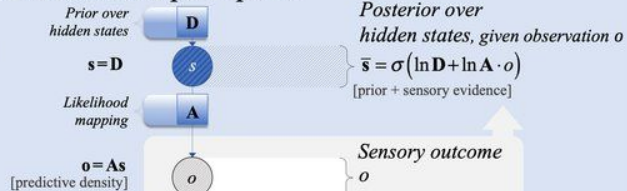
Steps of Increasing Model Complexity



Step 2: Anticipation

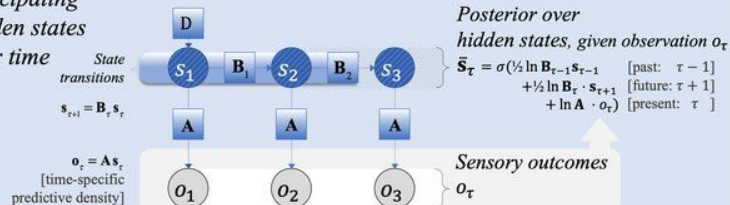
M₁. Generative model of perception

Perceiving
hidden states



M₂. Generative model of anticipation

Anticipating
hidden states
over time



2: A Generative Model of Anticipation (M₂ in Figure 1, Bottom Panel).

Generative Model (P)

$$P(s_1) = \underbrace{\text{Cat}(\mathbf{D})}_{\text{initial state prior}}$$

$$P(s_\tau | s_{\tau+1}) = \underbrace{\text{Cat}(\mathbf{B}_\tau)}_{\text{state transitions}}$$

$$\mathbf{s}_1 = \mathbf{D}$$

$$\underbrace{\mathbf{s}_{\tau+1}}_{\text{state expectations}} = \mathbf{B}_\tau \mathbf{s}_\tau$$

$$\underbrace{\mathbf{o}_\tau}_{\text{outcome expectations}} = \mathbf{A} \mathbf{s}_\tau$$

Approximate Posterior Beliefs (Q)

$$Q(s_\tau) = \underbrace{\text{Cat}(\bar{\mathbf{s}}_\tau)}_{\text{state posterior}}$$

$$\bar{\mathbf{s}}_1 = \sigma\left(\frac{1}{2} \ln \mathbf{D} \quad + \ln \mathbf{A} \cdot \mathbf{o}_1 + \frac{1}{2} \ln \mathbf{B}_1 \cdot \bar{\mathbf{s}}_2\right)$$

$$\bar{\mathbf{s}}_2 = \sigma\left(\frac{1}{2} \ln \mathbf{B}_1 \bar{\mathbf{s}}_1 + \ln \mathbf{A} \cdot \mathbf{o}_2 + \frac{1}{2} \ln \mathbf{B}_2 \cdot \bar{\mathbf{s}}_3\right)$$

$$\bar{\mathbf{s}}_3 = \sigma\left(\underbrace{\ln \mathbf{B}_2 \bar{\mathbf{s}}_2}_{\text{forward messages}} + \underbrace{\ln \mathbf{A} \cdot \mathbf{o}_3}_{\text{sensory evidence}} + \underbrace{\quad}_{\text{backward messages}}\right)$$

Forward Messages: Sensory evidence

Backward Messages: Predictive priors

Steps of Increasing Model Complexity



Step 3: Action

M₃. Generative model of action

Inferring phenotype-congruent policies from...

Baseline prior over policies
Phenotypic outcome preferences



Action model: Expected free energy
 $G_{\pi} = \sum_{\tau} [o_{\pi\tau} (\ln o_{\pi\tau} - C) - \text{diag}(A \cdot \ln A) \cdot s_{\pi\tau}]$
[phenotypic risk + perceptual ambiguity]

Prior over policies
 $\pi = \sigma(-E_{\pi} - G_{\pi})$
[baseline + action model]

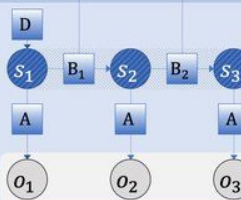
Posterior over policies
 $\bar{\pi} = \sigma(-E_{\pi} - G_{\pi} - F_{\pi})$
[priors + perceptual evidence]

Perceptual evidence: Free energy
 $F_{\pi} = \sum_{\tau} [s_{\pi\tau} \cdot (\ln s_{\pi\tau} - \ln A \cdot o_{\tau}) - \frac{1}{2} \ln B_{\pi\tau-1} \cdot s_{\pi\tau-1} - \frac{1}{2} \ln B_{\pi\tau+1} \cdot s_{\pi\tau+1}]$

(M₂ | π)

...perceived hidden states
 $s_{\pi\tau+1} = B_{\pi\tau} s_{\pi\tau}$
[policy-specific state priors]

$O_{\pi\tau} = A s_{\pi\tau}$
[policy-specific outcome priors]



Posterior over hidden states
 $\bar{s}_{\pi\tau} = \sigma(\frac{1}{2} \ln B_{\pi\tau-1} s_{\pi\tau-1} + \frac{1}{2} \ln B_{\pi\tau} \cdot s_{\pi\tau+1} + \ln A \cdot o_{\tau})$

Sensory outcomes
 o_{τ}

Policy Selection as Bayesian Model Selection

Choosing action sequences (policies) that represent potential versions of the future to achieve favorable outcomes.

Prior Policy Beliefs (π):

- **En:** Baseline Prior Expectation (habits/experience).
- **Gn:** Expected Free Energy (risk & ambiguity).
- **Phenotypic Priors:** Preferred sensory outcomes based on agent type.

How agents make decisions by evaluating different policies.

Prior Beliefs (Generative Model) (P)

$$P(\pi) = \underbrace{Cat(\pi)}$$

policy prior

$$\underbrace{\pi}_{\text{policy expectations}} = \sigma \left(\underbrace{-\mathbf{E}_\pi}_{\text{baseline prior}} - \underbrace{\mathbf{G}_\pi}_{\text{action model}} \right)$$

$$\underbrace{\mathbf{G}_\pi}_{\text{expected free energy}} = \sum_{\tau} \underbrace{\mathbf{o}_{\pi\tau} \cdot (\ln \mathbf{o}_{\pi\tau} - \mathbf{C})}_{\text{expected phenotypic risk}} - \underbrace{diag(\mathbf{A} \cdot \ln \mathbf{A}) \cdot \mathbf{s}_{\pi\tau}}_{\text{expected perceptual ambiguity}}$$

$$\underbrace{\mathbf{C}}_{\text{phenotypic preferences}} = \ln P(o_\tau)$$

$$P(s_{\tau+1} | s_\tau, \pi) = \underbrace{Cat(\mathbf{B}_{\pi\tau})}_{\text{policy-specific state transitions}}$$

$$\mathbf{s}_1 = \mathbf{D}$$

$$\mathbf{s}_{\pi\tau+1} = \mathbf{B}_{\pi\tau} \mathbf{s}_{\pi\tau}$$

$$\underbrace{\mathbf{o}_{\pi\tau}}_{\text{policy-specific expectations}} = \mathbf{A} \mathbf{s}_{\pi\tau}$$

Posterior Beliefs (Q) and Expectations

$$Q(\pi) = \underbrace{Cat(\bar{\pi})}$$

policy posterior

$$\bar{\pi} = \sigma \left(\ln \mathbf{E}_\pi - \mathbf{G}_\pi - \underbrace{\mathbf{F}_\pi}_{\text{perceptual evidence}} \right)$$

$$\mathbf{F}_\pi = \sum_{\tau} \bar{\mathbf{s}}_{\pi\tau} \cdot (\ln \bar{\mathbf{s}}_{\pi\tau} - \ln \mathbf{A} \cdot o_\tau - \underbrace{-\frac{1}{2} \ln \mathbf{B}_{\pi\tau-1} \bar{\mathbf{s}}_{\pi\tau-1} - \frac{1}{2} \ln \mathbf{B}_{\pi\tau} \cdot \bar{\mathbf{s}}_{\pi\tau+1}}_{\text{policy-specific prediction error}})$$

$$Q(s_\tau | \pi) = \underbrace{Cat(\bar{\mathbf{s}}_{\pi\tau})}_{\text{policy-specific state posterior}}$$

$$\bar{\mathbf{s}}_{\pi\tau} = \sigma \left(\frac{1}{2} \ln \mathbf{B}_{\pi\tau-1} \bar{\mathbf{s}}_{\pi\tau-1} + \ln \mathbf{A} \cdot o_\tau + \frac{1}{2} \ln \mathbf{B}_{\pi\tau} \cdot \bar{\mathbf{s}}_{\pi\tau+1} \right)$$

Steps of Increasing Model Complexity



Step 4: Implicit Metacognition

Internal Confidence & Expected Precision

The agent's ability to internally assess and update its confidence in its action model.

Expected Precision (γ)

An internal metric of the model's subjective fitness.

- Higher γ = Stronger influence of expected free energy on policy selection.
- Indicates the agent places more trust in its current model.

How agents modulate trust in their internal models.

M₄. Generative model of implicit metacognition

Estimating your confidence in...

Prior over precision

$$P(\gamma) = \Gamma(1, \beta)$$

Expected precision
 $\Rightarrow \gamma = 1/\beta$



Posterior over precision

$$Q(\gamma) = \Gamma(1, \bar{\beta}); \bar{\beta} = \beta - AC \\ \Rightarrow \bar{\gamma} = 1/\bar{\beta}$$

Affective charge

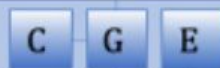
$$AC = (\pi - \bar{\pi}) \cdot G_{\pi}$$

$(M_3 | \gamma)$

...a phenotype-congruent model of actions on...

Prior over policies

$$\pi = \sigma(-E_{\pi} - \gamma G_{\pi})$$

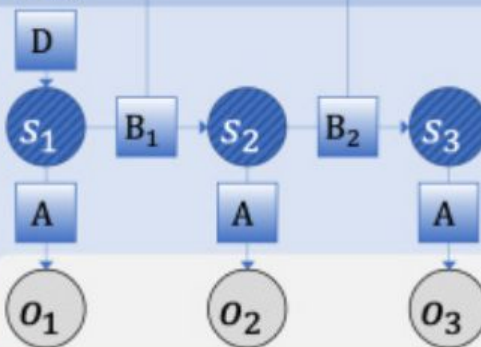


Posterior over policies

$$\bar{\pi} = \sigma(-E_{\pi} - \gamma G_{\pi} - F_{\pi})$$

$(M_2 | \pi)$

...perceived hidden states



Policy-specific posterior over hidden states
 $\bar{s}_{\pi\tau}$

Sensory outcomes
 o_{τ}

Prior Beliefs (Generative Model) (P)

$$P(\gamma) = \Gamma(1, \underbrace{\beta}_{\text{temperature parameter}})$$

$$\underbrace{\gamma}_{\text{expected precision}} \equiv E_{P(\gamma)}[\gamma] = 1/\beta$$

$$\pi = \sigma(-E_{\pi} - \underbrace{\gamma G_{\pi}}_{\text{precision-weighted action model}})$$

Posterior Beliefs (Q) and Expectations

$$Q(\gamma) = \Gamma(1, \bar{\beta})$$

$$\underbrace{\bar{\gamma}}_{\text{posterior precision}} \equiv E_{Q(\gamma)}[\gamma] = 1/\bar{\beta}$$

$$\bar{\beta} = \beta - AC$$

$$\underbrace{AC}_{\text{affective charge}} = \underbrace{(\pi - \bar{\pi}) \cdot G_{\pi}}_{\text{phenotypic progress}}$$

$$\bar{\pi} = \sigma(-E_{\pi} - \gamma G_{\pi} - F_{\pi})$$

Affective Charge (AC)

A measure of the discrepancy between an agent's **predicted** and **observed** outcomes under its current policy model.

Mathematical Definition

$$AC = (\pi - \vec{\pi}) \cdot G\pi$$

Where π represents current policy beliefs and $G\pi$ represents expected free energy.

Step 4: Implicit Metacognition

The active inference framework introduces **implicit metacognition**: a self-assessment mechanism for dynamic calibration.

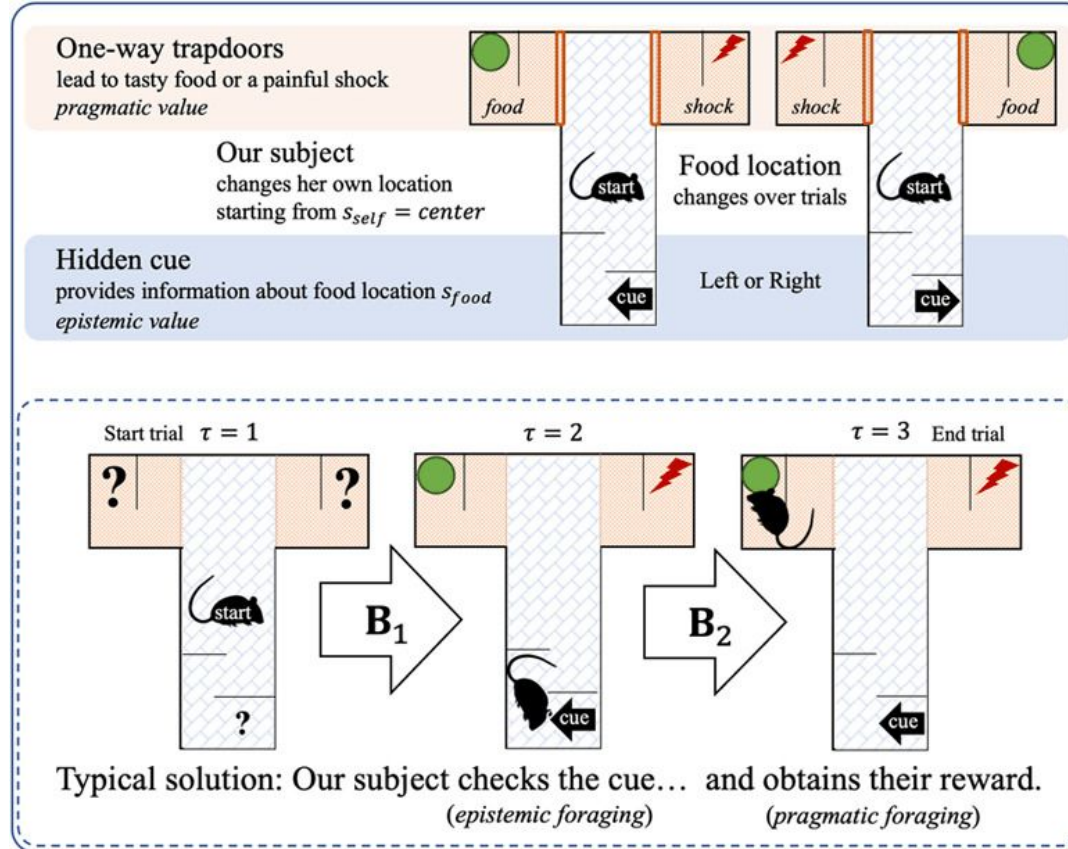
1. Belief Updating

The agent updates its internal beliefs about the state of the environment based on new sensory data.

2. Confidence Adjustment

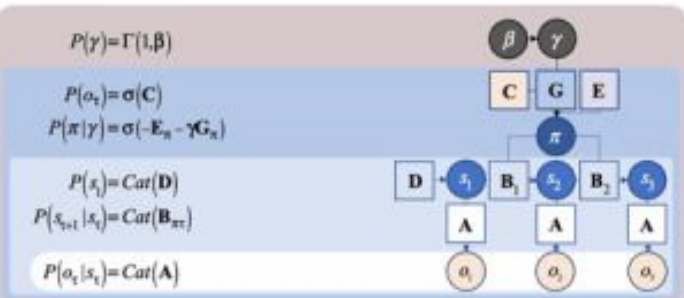
The agent adjusts confidence levels based on how well predictions match observed outcomes.

T-Maze Paradigm



Generative model

$$P(\gamma, \pi, \tilde{s}, \tilde{o}) = P(\gamma) \frac{P(\pi|\gamma)}{\beta \text{ C, G}_\pi \text{ E}_\pi} \frac{P(s_1)}{\mathbf{D}} \prod_{t=1} [P(s_{t+1}|s_t, \pi) \frac{P(o_t|s_t)}{\mathbf{A}}]$$



D Initial state prior

$$P(s_1)$$

$s_{\text{food}}: \{ \text{left}, \text{right} \}$
 $\mathbf{D}_{\text{food}} = [0.5, 0.5]$
 $s_{\text{self}}: \{ \text{center}, \text{left}, \text{right}, \text{cue} \}$
 $\mathbf{D}_{\text{self}} = [0.976, 0.008, 0.008, 0.008]$

C Preferences over outcomes

$$P(o_t)$$

$o_{\text{reward}}: \{ \text{none}, \text{green}, \text{red} \}$
 $\mathbf{C}_{\text{reward}} = [0, 4, -6]$
 $o_{\text{cue}}: \{ \text{none}, \text{left}, \text{right} \}$
 $\mathbf{C}_{\text{cue}} = [0, 0, 0]$

A Likelihood mapping – conditioned upon food and self

$$P(o_t|s_t)$$

$s_{\text{food}}: \text{left}, \text{right}$
 $s_{\text{self}}: \text{center}, \text{left}, \text{right}, \text{cue}$

o_{reward}
 $\mathbf{A}_{\text{reward}} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & .98 & .02 & 0 \\ 0 & .02 & .98 & 0 \end{bmatrix}$

o_{cue}
 $\mathbf{A}_{\text{cue}} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

B_t State transitions

$$P(s_{t+1}|s_t)$$

s_t
 s_{t+1}

$\mathbf{B}_{\text{center}} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $\mathbf{B}_{\text{left}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $\mathbf{B}_{\text{right}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $\mathbf{B}_{\text{cue}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

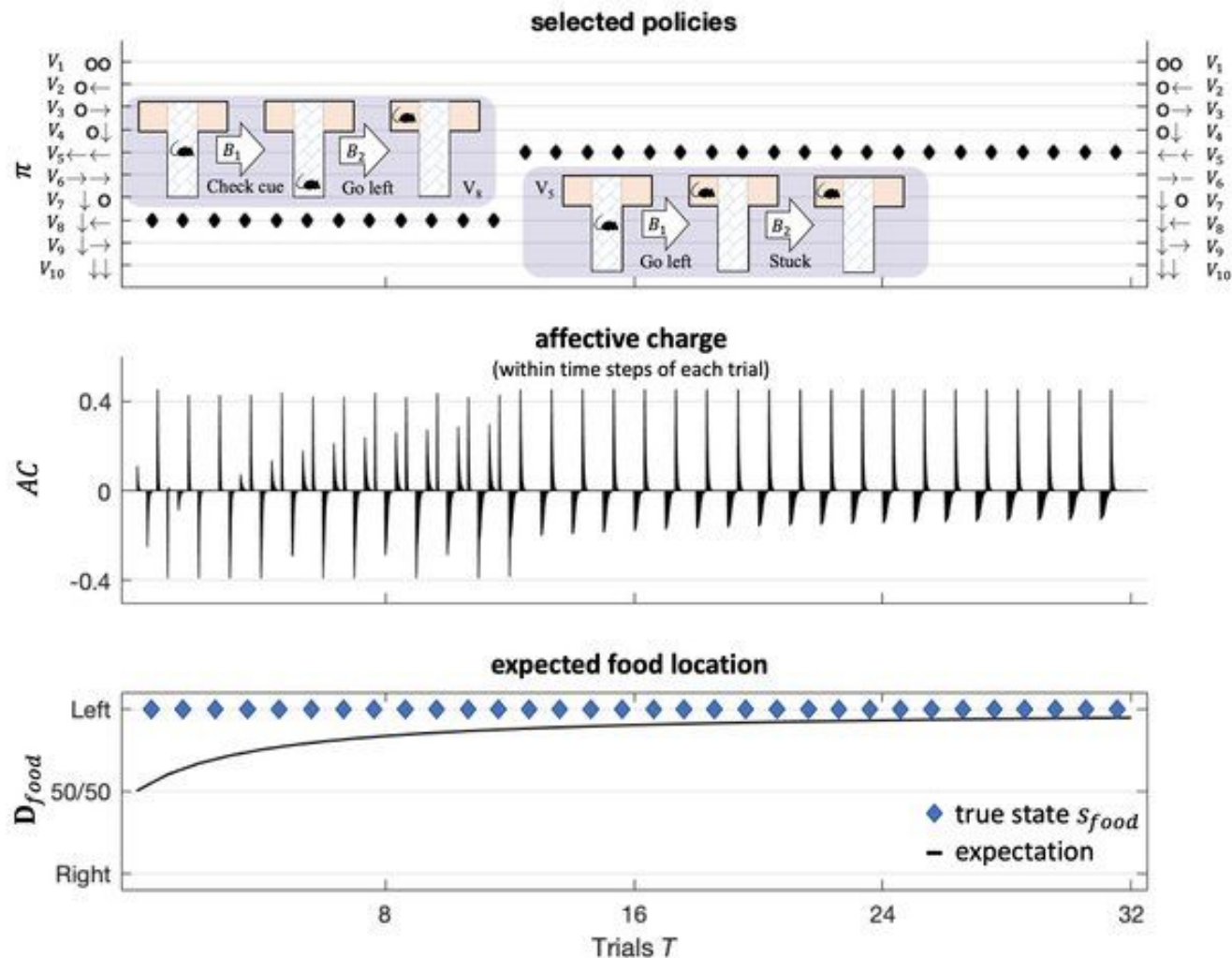
E_π Baseline prior over policies π: {V₁, V₂, ..., V₁₀}

Examples V₅, V₆, V₈, V₉

$\mathbf{V} = \begin{bmatrix} \text{B}_1 & \text{B}_2 \\ \vdots & \vdots \end{bmatrix}$

$\mathbf{E}_\pi = \begin{bmatrix} 2.3 \\ 2.3 \\ 2.3 \\ 2.3 \\ 2.3 \\ 2.3 \\ 2.3 \\ 2.3 \end{bmatrix}$

The Dirichlet parameterization is used to model the agent's beliefs about where the reward is located.



Agent Policy Selection & Behavior

Policies represent sets of actions taken within a T-maze environment, shifting from exploration to exploitation over time.

Visual Key:

Arrows: Movement towards left or right maze arms. | **Circles:** Staying in or returning to center.

Exploratory Behavior (Trials 1-12)

The agent checks the cue in the lower arm before making a choice.

- Reflects uncertainty about reward location.
- Decision based on cue indication.
- Higher computational cost for information gathering.

Exploitative Behavior (After Trial 12)

The agent moves directly to the left arm reward location.

- Reflects increased confidence in knowledge.
- No longer checks the cue.
- Optimized path for consistent reward retrieval.

Affective Valence and Model Fitness

Positive Valence (+)

Increases reliance on prior expectations.

- Greater confidence in the internal model.
- Reinforces current behavioral strategies.

Negative Valence (-)

Reduces reliance on priors.

- Reflects increased uncertainty.
- Prompts model updating or information gathering.

Risk Sensitivity: Influences how the agent balances exploration and exploitation based on affective state.

Deep Temporal Models

Exploring Hierarchical Structures in Affective Computing

Hierarchical Structure

Simulating complex inference through multi-layered temporal depth.

Generative Models & Affect

Inferring internal affective states based on environmental experience.

Key Mechanisms

- **Role of Generative Models:** Establish the framework for top-down predictions and error minimization.
- **Top-Down Predictions:** Drive the agent's expectations and influence sensory processing of affective cues.

Generative Model for Affective Inference

Simulating how agents infer affective states (positive or negative) based on environmental experiences through hierarchical temporal structures.

Level 1

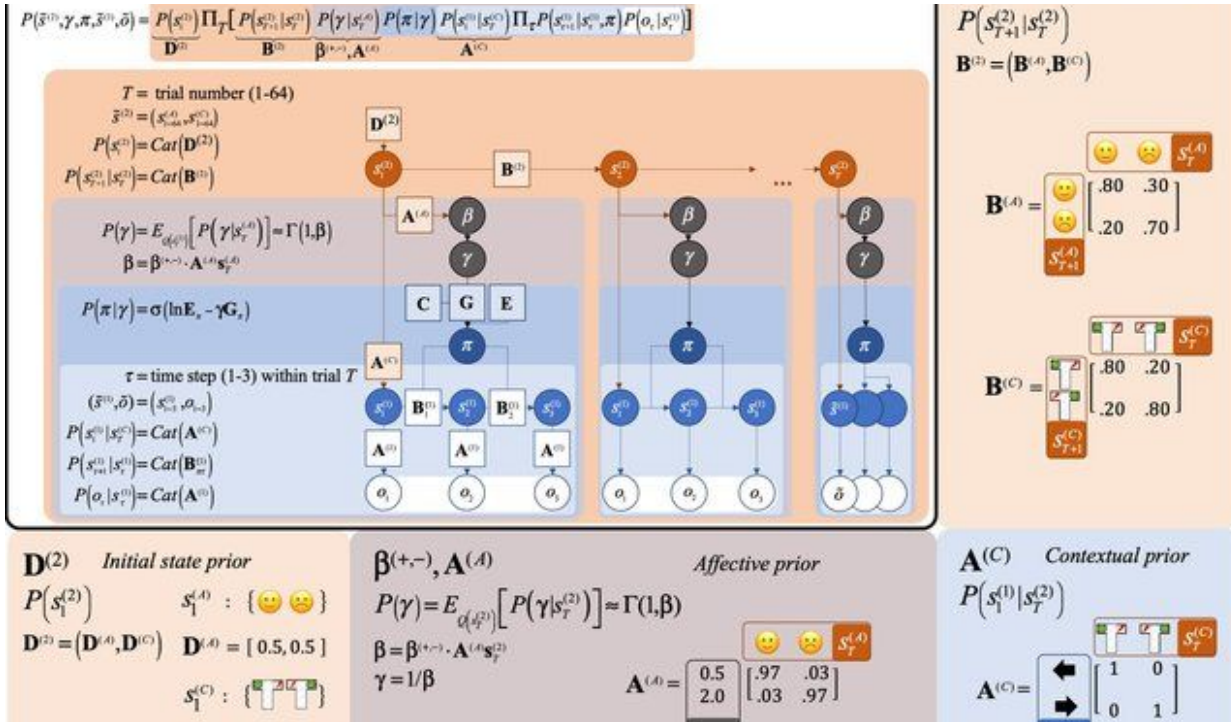
Affective State

Inference

Precision

Output

Action



Modeling Affective Valence in AI

Simulating emotional states for artificial agents to guide decision-making

Defining Affective Valence

A model for agents (robots/programs) to simulate emotional states like feeling "good" or "bad." This helps agents evaluate environmental success and guides decisions.

Charge vs. Valence

Affective Charge: Precision and confidence in beliefs.

Valence: Broader emotional evaluation influencing behavior.

Core Framework Benefits

Together, these elements contribute to a comprehensive framework for modeling affective states in artificial intelligence systems.

- Enables more nuanced cognitive process simulation.
- Provides mechanisms for motivation and behavioral influence.
- Improves agent adaptability in dynamic environments.

Theoretical Foundations

Evolution of affective valence modeling in neural and synthetic agents

Phaf & Rotteveel (2012)

Proposed valence as a **match-mismatch** process in neural networks, reflecting how well an architecture fits its inputs.

Joffily & Coricelli (2013)

Linked valence to the **rate of change in variational free energy**, measuring agent model vs. sensory input.

Current Approach

Introduces **affective charge** as a metric for changes in expected precision, reflecting model confidence.

This framework demonstrates how synthetic agents use emotional states to evaluate and improve environmental interaction.

- Integrates expected precision into higher-level state representations.
- Guides decision-making in complex, dynamic settings.
- Enhances AI through a computationally grounded emotional perspective.

Reference

Deeply felt affect: Understanding emotions through deep active inference

By Casper Hesp



Watch on YouTube: <https://www.youtube.com/watch?v=BLDqaERoZUQ>